Getting Started with Bioinformatics Data Analysis

How is this course different?

- Most bioinformatics classes/books focus on algorithms and implementation details.
- This course focuses on **information processing** and **data analysis**.
- We'll teach you **what to do** with data to extract the **information** contained inside it.
- Bioinformatics from a data oriented perspective

Flawed view of bioinformatics

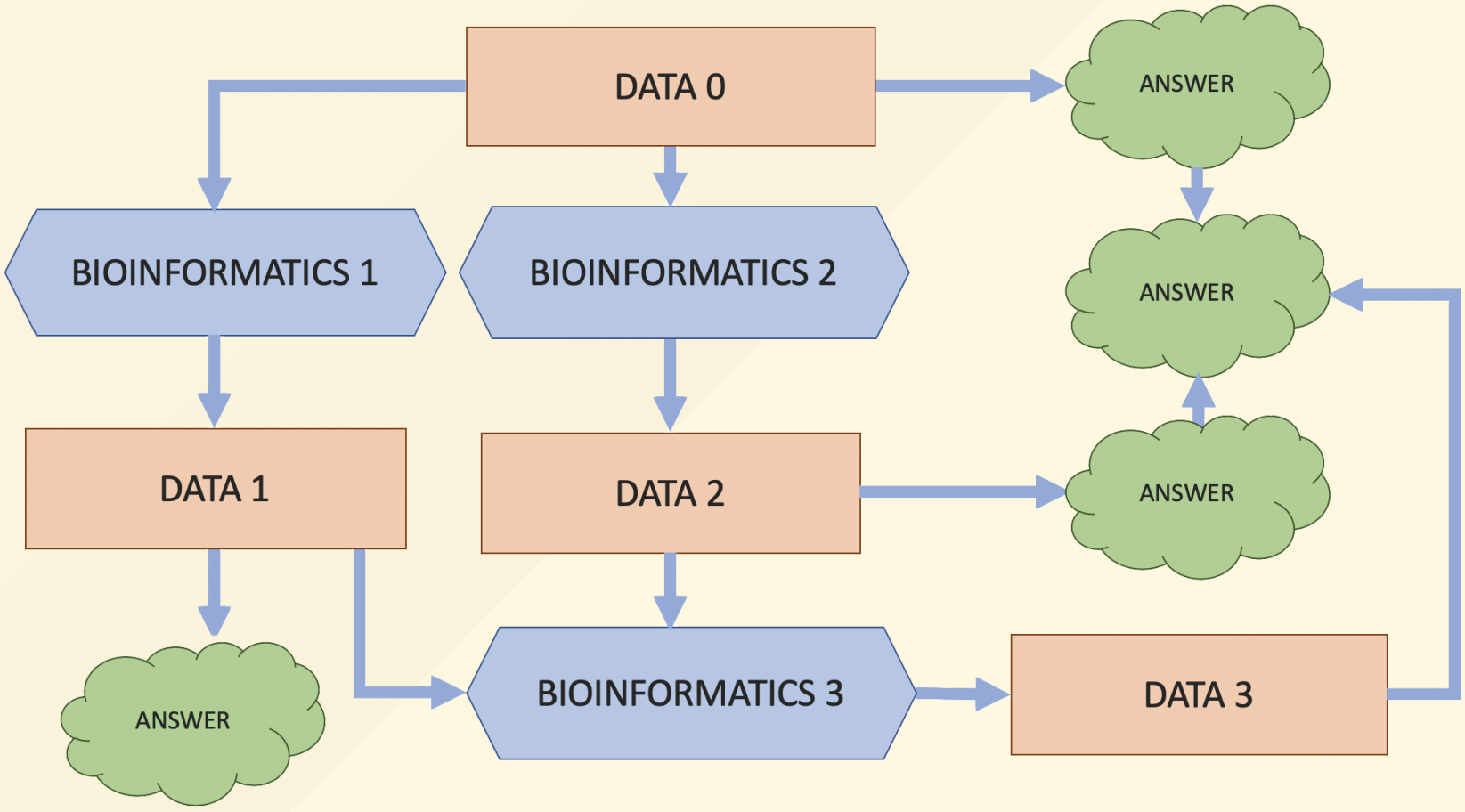
Most biologists believe bioinformatics works the following way



Misconception: *scientists need to find the right tool that gives them the answer*

This world view is suprisingly pervasive.

Bioinformatics in reality



Tools create new data. Data may have some answers.

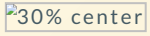
IN BIOINFORMATICS

DATA IS KING!



Course Materials

- **Biostar Handbook, 2nd Edition**
- Course: **Bioinformatics Data Analysis**



Rationale the course

- Life sciences have become a **data driven** science.
- Data are represented as **text based** files in various formats.
- What is inside the data? More than what we know how to extract.

Course requirements

A computer and a "can-do" attitude 😄

You may use `MacOS`, `Linux` or `Windows 10`

This is a good time to tell your yourself/advisor/mom/boss/SO that this course recommends that you get a Mac.

Tell them I said that a new Macbook will make you incredibly productive 👩 👸 .

Lecture format

We will study well defined concepts at a time.

- In each lecture we try to cover one topic.
- First background information.
- Practical examples that tie in with the topic
- Class exercises + homework.

Assignments build on the lecture. You will need to **repeat** and **redo** what was done in the lecture.

Course motto

> **What I cannot create, I do not understand.**
> **Richard Feynman**

Bioinformatics is very different from typical Life Sciences.

It is not a "descriptive" science where you need to learn various "ground truths".

It is a information science - you need to decode information locked away within data.

You have to learn how to do it yourself.

Complexity vs decision making

Bioinformatics analyses require a *large* number of *relatively simple* decisions.

Most of which need to be correct! That's what makes it difficult!

There are no *absolute rules*, only guidelines. You must learn to improvise and adapt.

That what this course is about: *How to not be afraid of making decisions.*

Expectations

You can learn only by doing it!

Spend a few hours on each lecture beyond what we show:

- Explore behaviors.
- Expand the scope of the study.
- Try new solutions, push the boundaries.

Time flies when you know what you are doing.

Bioinformatics Today

Combination of different sciences

1. **Information technology:** data storage, transfer, data transformation
2. **Computer science:** algorithms advanced data structures, software tools
3. **Statistics** yet traditional statistics is not well suited for modeling systematic errors over large number of observations
4. **Life Sciences:** biological hypothesis testing, interpretation

How is bioinformatics practiced?

1. **Command line tools.** "Action words" are chained together to form a pipeline. Data “flows” from one command to the other.
2. **R Programming environment.** A high level programming statistical environment. Best suited for later stages of analyes.
3. **Tools with graphical user interfaces**: Web based interfaces to command line tools (Galaxy), large selection of commercial software.

What are command line tools like?

"Action words" that eventually become familiar:

```
bwa mem read1.fq read2fq | samtools sort > alignment.bam
```

"Chained" with characters such as `|` and `>` to form a pipeline. Data **flows** from one command into the other.

Resembles natural language. Provides generic building blocks. Adaptive and expressive. Easy to repeat or share the same command.

There is a substantial learning curve to it.

R Programming environment

Specifically the Bioconductor package in R.

A high level statistical programming environment.

Attempts to provide with “simple” constructs that perform complex tasks. Excellent visualization capabilities

Unfortunately as a programming language R is not well designed. It is quite challenging (maddening?) to use it correctly.

What does an analysis look like in R

Provides more "specialized" action words:

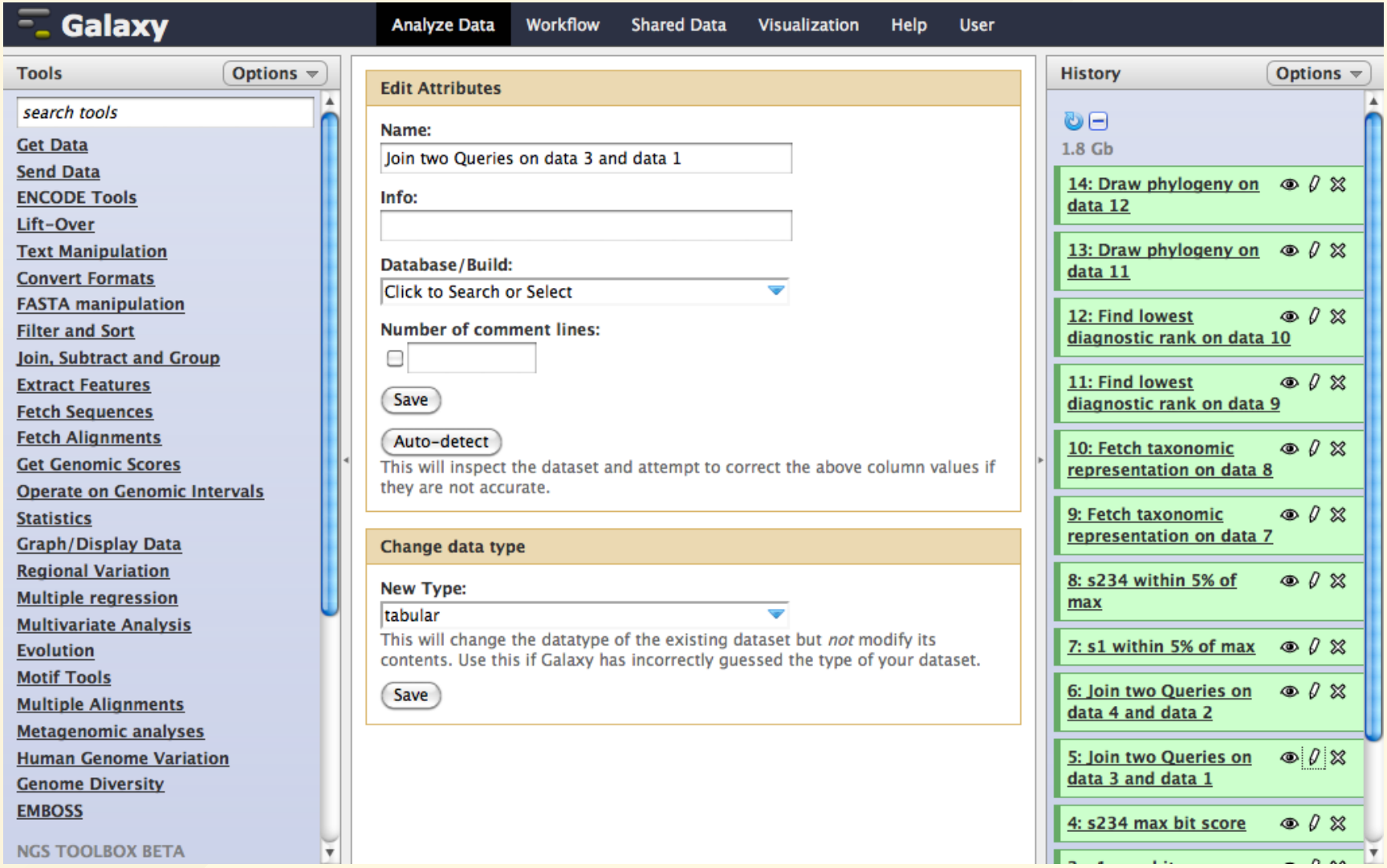
```
biocLite("DESeq")
library(DESeq)
count = read.table("stdin", header=TRUE, row.names=1 )
cond1 = c("control", "control", "control")
cond2 = c("treatment", "treatment", "treatment")
conds = factor(c(cond1, cond2))
cdata = newCountDataSet(count, conds)
esize = estimateSizeFactors(cdata)
edisp = estimateDispersions(esize)
rdata = nbinomTest(edisp, "control", "treatment")
```

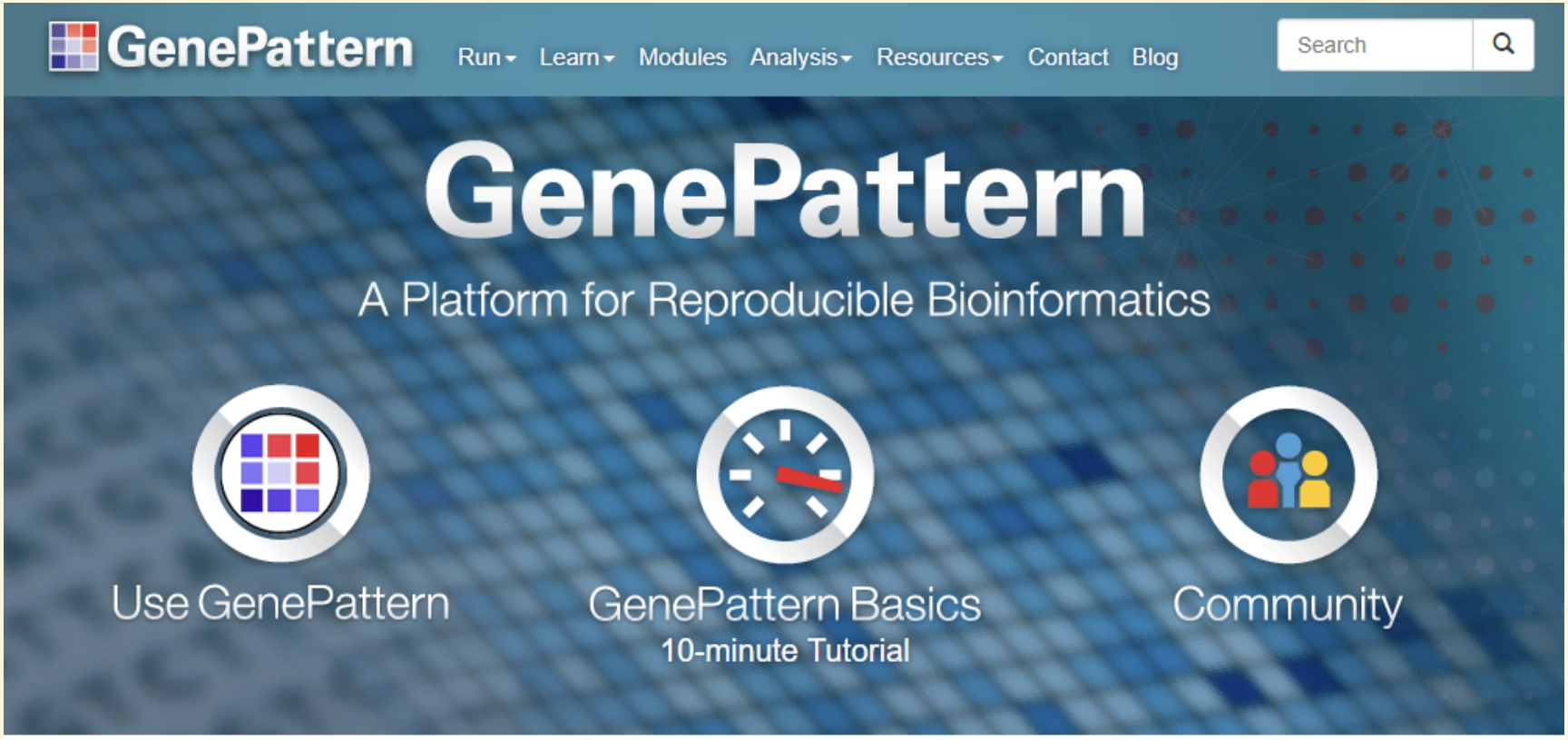
Negative: requires lots of "book-keeping". Exceedingly **easy to make mistakes** (mix up labels etc.) that are hard to notice.

How do grapical user interfaces work?

- So called "discoverable" environment that initially appears to be simple. This "simplicity" is deceiving.
- Often lag far behind when it comes to applying the newest of analyses.
- Difficult to understand what has been done to a given dataset.
- *"I have analyzed my data with Galaxy"* what does that even mean?
- Difficult to repeat the process the same way.

What does the Galaxy interface look like?





Features

Powerful genomics tools in a user-friendly interface

GenePattern provides hundreds of analytical tools for the analysis of gene expression (RNA-seq and microarray), sequence variation and copy number, proteomic, flow cytometry, and network analysis. These tools are all available through a Web interface with no programming experience required.

Blog > GP updates

- Scheduled Maintenance Downtime - Saturday, July 22, 2017
- End of support for GISTICPreprocess
- End of support for GENE-E
- GenePattern 3.9.10 is released!
- GenePattern Notebook Repository Update Released - v17.05

view more >

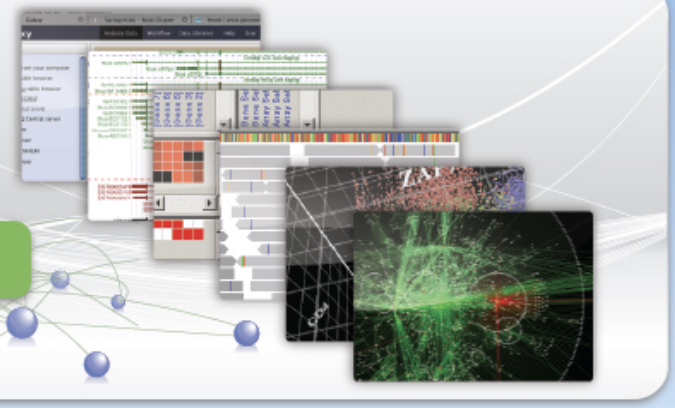
GENOMESPACE

User Login

What is GenomeSpace? Tools Recipes Documentation Developers Support About

GENOMESPACE

Frictionless connection of bioinformatics tools



Register User Login

STATUS

04.11.17 09:09PM Currently all systems are operating normally.

Calendar of Upcoming Events

Citing GenomeSpace

To cite your use of GenomeSpace, please reference Qu K, Garamszegi S, Wu F, et al. Nature Methods. 2016 Jan 18. doi: 10.1038/nmeth.3732.

F1000 Research

Check out our F1000 GenomeSpace Channel for published, community-contributed recipes.

WHAT'S NEW

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OCLC > Public domain > List of open-source bioinformatics software > Genomespace > Computing platform > PlayStation > Tiny Pop > QVC Beauty

Aug 13, 2017

Vera Matser

What do most "graphical" tools do behind the scenes?

"Graphical" tools create and execute command line tools.

Why not run your own commands directly?

Tools vs data revisited

Modern instruments produce immense amounts of data.

Impossible to interpret them without various tools.

For many people "knowing tools" became the bioinformatics skill.

Bioinformatics skill means understanding how to extract information from data.

Tools change all the time - we can learn more from the same data.

What is more valuable skill

Both are "bioinformatics tasks"

Run the `bowtie` aligner, then run the `cuffdiff` software.

Create a spliced alignment file, then quantify the abundances by intersecting the alignments with the genomic intervals, then apply a statistical test to select differentially expressed entries.

Tools change. Concepts don't. Over time tools implement the same concepts better and better.

Where to go next?

1. Read chapters 1 and 2
2. Set up your computer as instructed in chapter 2

Installation may pose a few unexpected challenges - system updates, passwords, downloading lots of files etc.

Get to it right away as it may take you a while to set everything up.

💻 The Biostar Handbook has detailed instructions on setting up your computer.

⚠️ Set up your computer!

You need a properly set up computer to complete the assignments.